

EXPERIMENT 6

Objective: To implement Inorder, Preorder and Postorder on a given Binary Search Tree.

```
#include <bits/stdc++.h>
using namespace std;
```

```
class Node {
public:
    int data;
    Node* left;
    Node* right;
```

```
Node(int val) {
    data = val;
    left = nullptr;
    right = nullptr;
}
```

```
void printPreorder(Node* node) {
    if (node == nullptr) {
        return;
    }
    cout << node->data << " ";
    printPreorder(node->left);
    printPreorder(node->right);
}
```

```
void printInorder(Node* node) {
    if (node == nullptr) {
        return;
    }
    printInorder(node->left);
    cout << node->data << " ";
    printInorder(node->right);
}
```

```

    }

    void printPostorder(Node* node) {
        if (node == nullptr) {
            return;
        }
        printPostorder(node->left);
        printPostorder(node->right);
        cout << node->data << " ";
    }
};

int main() {
    Node* root = new Node(100);
    root->left = new Node(20);
    root->right = new Node(200);
    root->left->left = new Node(10);
    root->left->right = new Node(30);
    root->right->left = new Node(150);
    root->right->right = new Node(300);

    cout << "Preorder Traversal: ";
    root->printPreorder(root);

    cout << endl;
    cout << "Inorder Traversal: ";
    root->printInorder(root);

    cout << endl;
    cout << "Postorder Traversal: ";
    root->printPostorder(root);
}

```

Output:

```

PS C:\Second Year\3rd Semester\Advanced DSA using C++> cd "c:\Sec
$?) { g++ Order_finder.cpp -o Order_finder } ; if ($?) { .\Order_
Preorder Traversal: 100 20 10 30 200 150 300
Inorder Traversal: 10 20 30 100 150 200 300
Postorder Traversal: 10 30 20 150 300 200 100
PS C:\Second Year\3rd Semester\Advanced DSA using C++\BST>

```